

RUI HENG FOO

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EDUCATION

Master's Degree in Computer Science | Arizona State University

Expected graduation: May 2026

- 4.0 CGPA
- Relevant Coursework:** Software Engineering, Semantic Web Mining, Software Quality Assurance and Testing, Software Project, Process and Quality Management

Bachelor's Degree in Computer Science | Arizona State University

- Dean's List – Spring 2025, Fall 2024, Spring 2024, Fall 2023, Spring 2023, Fall 2022
- Relevant Coursework:** Distributed Software Development, Operating Systems, Principles of Programming Languages, Information Assurance, Design and Analysis of Algorithms, Theoretical Computer Science, Data Mining

WORK EXPERIENCE

Research Aide | Data Mining and Machine Learning Lab | SCAI ASU

June 2025 – Present

- Lead developer** for Digital Twin for Disasters, building an extensible wildfire digital twin platform with **Python** backend services for a **Vite.js** frontend, stochastic simulation pipelines, and cloud-based geospatial data integration.
- Lead developer** of Digital Twin for Battlefield, delivering a **Python** and **React.js** based digital twin system by extending and stabilizing an open-source codebase (*Panopticon AI*).
- Co-developed** a research application for a project with a paper accepted at **CIKM'25** (The 33rd ACM International Conference on Information and Knowledge Management).
- Collaborated closely with five PhD researchers in a fast-paced, sprint-based (**Agile**) development environment.
- Maintained clean, well-structured codebases, ensuring effective version control, modularity, and development workflows across projects.

Technology Support Assistant | School of International Letters and Cultures

October 2023 – May 2025

- Supported technology operations for 80+ classes, ensuring reliable delivery of services for 3,000+ students in a fast-paced academic environment.
- Coordinated with faculty and staff to resolve technical issues efficiently and maintain smooth workflows.

LAB EXPERIENCE

Undergraduate Research Volunteer | Data Mining and Machine Learning Lab

January 2025 – June 2025

- Developed backend-supported visual interfaces using Python Flask and Jinja to improve clarity and reliability of disaster assessment outputs.
- Implemented geospatial analysis workflows in Google Earth Engine using JavaScript for pixel-based disaster damage assessment.
- Investigated and evaluated data-driven approaches for disaster-related information extraction to support downstream analysis.

PROJECTS

Panopticon AI | React.js, Python

June 2025 – Present

- Open-source contributor to Panopticon AI, contributing bug fixes, system improvements, and issue development to enhance platform stability and usability.
- Investigated and resolved codebase issues, gaining hands-on experience with the project's architecture and workflows.
- Engaged with a project co-founder to discuss the platform's roadmap and identify opportunities for deeper technical contributions.

Stock Market Sentiment Analyzer | Python, REST APIs

October 2024 – November 2024

Project Lead

- Designed and led the development of a sentiment analysis tool that advises users on investment actions and delivers actionable insights based on real-time stock market sentiment.
- Leveraged **Python APIs** such as AlphaVantage, PRAW, and Textblob to pull stock market data, Reddit posts, and aggregated sentiment scores into an analysis pipeline.
- Implemented Backtrader, a robust backtesting framework to validate investment strategies, enabling data-driven decision-making.

Intro to distributed software development | ASP.NET, REST APIs, C#

January 2024 – April 2024

Project Lead

- Led the development of a **multi-layered backend web application** using **ASP.NET**, implementing clear separation between presentation, service, and data layers.
- Designed and implemented **RESTful web services** to support secure and modular client-server communication.
- Implemented user authentication, session management, and encrypted credential storage, ensuring secure access control.
- Coordinated deployment of backend services on the ASU WebStrar server, ensuring reliable integration across application components.

TECHNICAL SKILLS

- Languages:** Python, Java, C#, C/C++, SQL, HTML, JavaScript, CSS
- Frameworks/Libraries:** Flask, ASP.NET, Jinja, NumPy, Pandas, TensorFlow, Scikit-learn, NetworkX, Backtrader, REST APIs
- DevOps & Tools:** Google Cloud Storage (GCS), Google Earth Engine (GEE), Geospatial data processing (GeoTIFF, raster data), Git, Agile, Scrum, JUnit, Jupyter Notebook, Visual Studio, VS Code